An Advanced Electronic Health Record for Prescription Analysis Using YOLO and OCR with Spell Correction

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***Abstract*— Keeping accurate and well-organized medical records is still quite difficult in today's healthcare setting, particularly when it comes to handwritten prescriptions. In order to solve this problem, RxVision created a sophisticated technology that automatically transforms handwritten prescriptions into organized Electronic Health Records (EHRs). The system uses OCR for text extraction following utilizing the YOLO model to identify important sections of interest, such as medication names, dosing amounts, and usage instructions. A spell correction module improves OCR outputs to increase accuracy and reduce handwritten text interpretation problems. With a mAP@0.5 of 0.93, Precision of 94.1%, Recall of 91.3%, and an F1-score of 0.92, the final integrated model demonstrated its dependability and effectiveness with outstanding performance. Furthermore, the structured prescription data generated by the system enables downstream analytical tasks such as risk pattern identification and trend analysis, supporting clinicians in data-driven decision-making. All things considered, RxVision greatly enhances healthcare data management by making patient record maintenance and utilization quicker, more accurate, and simpler for practitioners and hospitals. This system way more managing the health records easier, more accurate, and faster, reducing the manual work for the hospitals and helping the doctors to make better and data-driven decisions**

Keywords*— Handwritten Prescription Recognition, Electronic Health Records (EHR), Optical Character Recognition (OCR), YOLO Object Detection, Medical Document

I. INTRODUCTION

In our modern civilization, several individuals do not care about their health, as they do not take the checkups regularly and do not want to see the doctor. Thus, to resolve this problem, we are planning to create an application that will allow tracking the health of a person without having to make a complete body scan. We are working towards medical prescription analysis with the help of Optical Character Recognition (OCR). Within it, patients will post the images of their prescriptions that will be analyzed and stored on a digital level. It will be an application that keeps on updating the medical records of all users when they have new prescriptions and hence retains a well-organized health profile in the long run. And the traditional ways find it more difficult to derive finer details out of the prescriptions and they do not sort out medicine-related information in a senseful manner. To overcome these situations, we propose a personalized Electronic Health Record (EHR) system that offers a private and accessible digital record for managing patient's medical histories. Our study integrates a Region of Interest (ROI) detection model and built on the YOLOv11 framework, combined with Tesseract OCR (version 5.5.1). The ROI and YOLO modules first identify the relevant sections of the prescription image and then the OCR extracts the text and converts it into a structured digital format. This gives the accurate recognition of prescription details and efficient storage of health data for long-term use and analytical decision support based on structured historical prescription data.Several studies have explored automated prescription analysis using optical character recognition (OCR) and deep learning techniques to digitize handwritten medical prescriptions [1], [7], [8].

Object detection–based approaches have also been applied to localize medicine-related regions from prescription images with promising accuracy [2]. Although these methods demonstrate significant progress, most existing systems focus primarily on partial information extraction, such as medication name recognition, and do not structure complete prescription details including dosage, frequency, and usage instructions. Furthermore, many current solutions rely on centralized data collection and processing, which raises privacy and security concerns when handling sensitive patient health information [14], [16]. These limitations motivate the need for a comprehensive, privacy-aware, and end-to-end prescription digitization framework, which is addressed in this work.In spite of some studies examining automated prescription analysis through the use of Optical Character Recognition (OCR) and deep learning methods, the majority of the current methods consider only some part of the prescription digitization pipeline.

The med name recognition or text extraction is produced in earlier work although the entire structuring of dosage, frequency and usage instructions needed to facilitate smooth Electronic Health Records (EHR) integration is not ensured. Additionally, the ability to survive a wide range of handwritings, spelling inaccuracies, and low quality image is also a major challenge. The current systems do not also have a single framework to integrate the proper region localization, text correction, and structured data creation to conduct long

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term clinical analysis. These shortcomings trigger the necessity to develop an end-to-end prescription digitization system, which is able to extract, correct, structure, and store accurately the prescription data in a form that can be used in the real-world healthcare implementation.The suggested RxVision model is capable of solving such issues by incorporating ROI identification, OCR improvement, and formatted EHR creation.

II. Literature survey

A. Prescription Recognition and Optical Character Recognition (OCR)

The accuracy of the extracting prescription information is critical in our modern healthcare, as the manual entry often leads to errors due to the poor handwriting of doctor or illegible notes [1]. This problem is taken care of by automated techniques, especially the Optical Character Recognition (OCR). The OCR accuracy is enhanced by a significant percentage of preprocessing methods, such as resizing, grayscale conversion, thresholding, and removal of noise to make text clearer [1], [7]. Normal expressions are commonly applied on the extraction of the structure data which consists of names of the patients, medications, dosages, and instructions and thus it is appropriate to store and analyze it [1]. OCR is used together with deep learning in several ways. One such example is the results of CNNs with preprocessing steps which show approximately 70 percent of accuracy on actual prescriptions [7], or transformer-based models such as TrOCR with Roboflow which can achieve up to 81.3 percent accuracy, this makes messy handwriting and dosage instructions practically manageable [8]. Also, the Natural Language Processing (NLP) has been used to classify prescriptions. Carchiolo et al. [2] used OCR, the EAST detector and Tesseract to read poor-quality scans, used spelling correction with Damerau-Levenshtein distance and rule-based NLP to categorize prescriptions. The experiment with more than 800,000 samples indicated that all of it could be automated with only 5% needing a manual examination.

B. AI in Drug Discovery and Prescription Improvement

AI has been increasingly applied in drug discovery and rational prescription management. Zhu [4] highlighted that large datasets, such as PubChem and ChEMBL, enable AI and deep-learning models to deal with the complex drug interactions and safetyanalytical decision support based on structured historical prescription data more effectively than traditional QSAR models. Jiménez-Luna et al. [6] noted that deep learning supports QSAR/QSPR modeling, de novo drug design, and synthesis planning but faces challenges such as high computational costs and “black-box” interpretability. Hybrid AI methods and explainable AI (XAI) approaches are suggested to enhance reliability and transparency.

AI is also being applied to improve antibiotic prescribing. Amin et al. [5] reviewed supervised machine learning models, including logistic regression, random forest, gradient boosting, SVM, and KNN, showing that these can reduce inappropriate antibiotic use.However, none of the studies incorporated clinician involvement or assessed usability in real hospital environments.

C. Clinical Decision Support and Error Reduction

Machine learning-driven clinical decision support systems (CDSS) are increasingly being used to reduce prescription errors. Segal et al. [15] demonstrated that probabilistic ML-based CDSS in inpatient settings reduced false alerts, improved alert usefulness, and influenced nearly 43% of prescription changes. Similarly, Rozenblum et al. [21] evaluated MedAware, which detected over 10,000 alerts from 748,000 patient records, achieving about 80% true positives and saving costs around $1.3M over five years. Wu et al. [3] tested an AI-assisted prescription review system in China, lowering irrational prescriptions from 27.7% to 24.1% and high-risk cases from 13.6% to 1.5%, demonstrating both clinical and economic benefits.

Pharmacy is another field that AI models are applied to in drug consultation, prescription checks, and case studies [11]. These models can improve efficiency, yet it is needed to be controlled by human means.

D. EHR Integration and Explainable AI

The integration of Electronic Health Records (EHR) with AI has allowed making medical analysis more personal and precise. Research indicates that EHR can be used to predict diseases such as diabetes, cardiovascular risk, and cancer using AI models [9], [14], [20]. The explainable AI (XAI) has been used to enhance the options of models. Payrovnaziri et al. [12] and Carrasco-Ribelles et al. [13] defined the following important strategies of XAI: rule-based, attention mechanisms, and feature importance, but there are still issues of reproducibility, trust, and real-world implementation.

Tsai et al. [20] showed that AI models trained on EHR data have superior results compared to traditional risk scores in cardiovascular prediction as well as offer explainable predictions, which results in clinician trust. Another aspect that Akhtar [19] emphasized the AI systems that value patient care in hospitals based on the principles of EHR and workflow management, demonstrating better efficiency and patient safety.

E. AI in Traditional and Personalized Medicine

There is also the application of AI in traditional medicine and clinical pharmacy. Park et al. [10] came up with a tool to examine decision-making in a one-on-one diagnosis and prescriptions, which showed discrepancies between stated and enacted clinical decisions. Li et al. [11] compared the generative AI systems and pharmacy activities, and their study concluded that even though some AI systems may work well, there is a necessity to control human actions and prevent mistakes. Zargoush et al. [18] used predictive-prescriptive analytics in the management of diabetes, and they showed great accuracy in the personalized prescription of medicines, but the difficulties of cost to compute and interpret the model persist.

III. RESEARCH GAP

Based on the literature review, it is noted that the majority of available strategies to digitize the prescription offer are mostly limited to single steps of medication name recognition through OCR or deep-learning methods. Most of the systems are based on manual image annotation, binarization, or centralized data processing, which make

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them less scalable and privacy-hazardous in healthcare settings. Moreover, a number of studies fail to capture all the prescription data including dosage, frequency, and instructions and fail to format the data retrieved in a way that can be added immediately into the Electronic health record (EHR) systems. Moreover, resistance to various handwriting styles and spelling variability is also an issue, and the model training that is privacy-preserving is hardly discussed.

In order to seal these gaps, the proposed RxVision system offers an end-to-end platform that combines YOLO-based region-of-interest detection, OCR-based text extraction, and spell correction in a bid to accurately computerize handwritten prescriptions into formatted EHR-readable data. The proposed approach, in contrast to the existing ones, focuses on full prescription structuring, enhanced resistance to handwriting differences, and enhancement of model privacy via federated learning. This directly addresses the found limitations since it allows scalable, secure, and reliable prescription digitization that is applicable to real-world healthcare implementation.

IV. PROPOSED SYSTEM

The said-system which is suggested to perform prescription analysis will combine the next level of the image processing, Optical-Character Recognition (OCR), and Deep Learning methods in order to identify and analyze the essential medical data on the hand written or printed prescriptions. The modules of the system have a certain type of work, which leads to the conversion of unstructured image data to structured machine-readable data applicable in healthcare analytics and electronic health records.

A. Data Ingestion

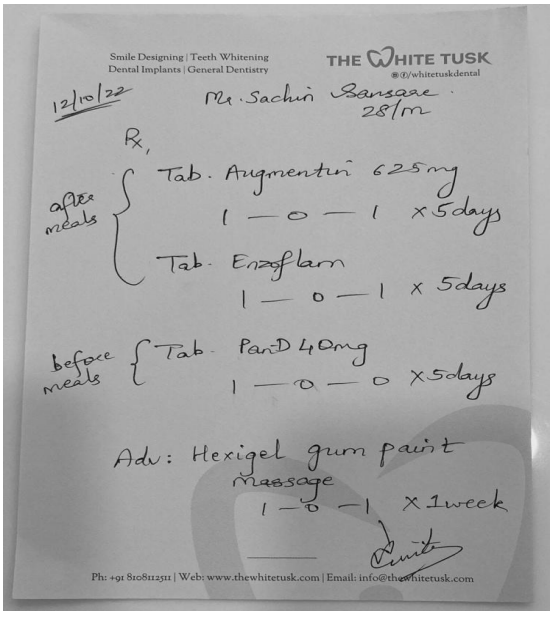


Fig.1. Sample Prescription image

The data ingestion phase serves as the entry point of the system and the sample prescription is shown in Fig. 1. The application interface allows a user to upload a prescription image and it is processed by the central server. To preserve the integrity and privacy of the data, the server gives this picture a temporary store in a safe setting. This step also highlights the importance of effective management of medical information because, in most cases, a prescription may hold sensitive information about a patient. The uploaded pictures are authenticated to verify that they are compatible and clear which means that only the supported formats (e.g., JPEG, PNG, or TIFF) will be forwarded. This will ensure that the later stages of processing are done with valid and quality information, and that errors in recognizing and analyzing are reduced.

B. Image Preprocessing and ROI Detection

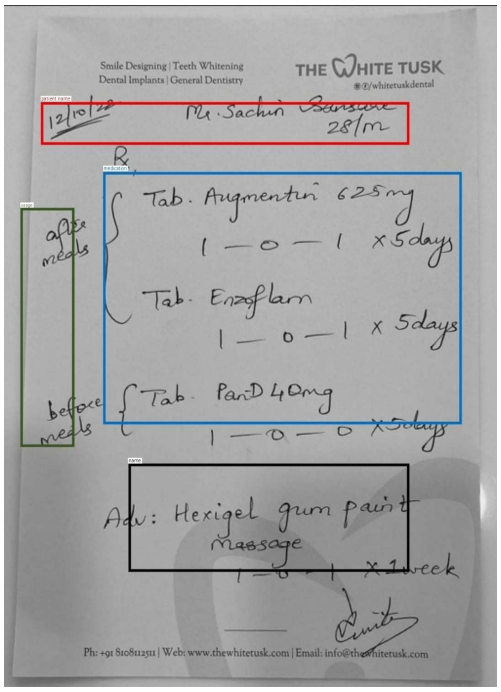


Fig.2. ROI detection

After the successful processing of the image, it is subject to certain preprocessing processes that is meant to improve its visual properties and enable it to be more presentable to the optical analysis. Some of the typical actions are size reduction of the image to the predetermined size, scaling down the image to grayscale and the use of noise reduction filters to remove the undesired artifacts or deformities which may cause distortion of text and thus interfere with text recognition. After preprocessing, a YOLOv11-based object detection model is employed to identify the *Regions of Interest (ROI)* within the image as shown in Fig. 2. These regions include the areas containing handwritten or printed prescription text, such as patient details, doctor's name, and medicine list. The YOLOv11 model is selected for its high detection speed and accuracy, making it ideal for real-time applications. The detected ROIs are then cropped and passed on for textual extraction, ensuring that only relevant parts of the image are processed by the OCR engine.

Fig.3 illustrates the detection accuracy of the YOLO-based model in identifying prescription regions of interest, demonstrating improved localization performance after model enhancement.

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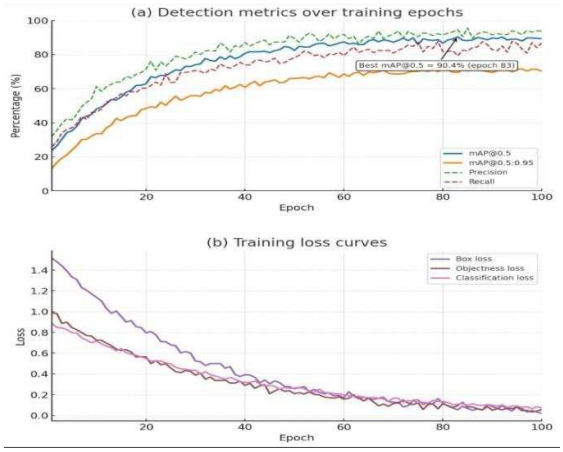


Fig.3. Performance of YOLO detection model

C. OCR Using Tesseract

The text recognition stage utilizes Tesseract OCR (version 5.5.1) which is an open source engine that is known to be the most successful when it comes to text recognition of both printed and handwritten text. The already processed ROI images are introduced to the Tesseract engine that uses feature extraction algorithms and pattern recognition algorithms to transform the information represented by pixels into edit and searchable textWhen handwritten prescriptions are used, the performance of OCR is usually affected by a variation in writing styles and spacing. To address all these, adaptive thresholding and line segmentation are used to overcome these difficulties before the execution takes place of OCR. And now the result text will construct the first digital expression of the content of the prescription upon which further correction and standardization will be done.

D. Error Detection and Terminology Mapping

Since the inherent variability of handwriting is given, and OCR the text we extracted is limited, an error detection and correction algorithm with a detection rate of 96% is used to minimize errors This is especially important when dealing with medical terms, where even a small misspelling may result in a medical error, and we also further apply the terminology mapping which will match the ROI boxes , e.g., abbreviation or informal writing("Paracetmol" to "Paracetamol") are mapped to identify the terms.

This ensures that names, dosages and units of medication used in various prescriptions remain the same, which increases the degree of accuracy of the downstream analysis.

E. Structured Data Creation

Once we have cleaned and standardized the data, it will be structured into a systemated format that can be used to store in database. The data schema is structured to contain the most important attributes which are patient information, information about the doctor, medication name, dosage, and frequency. Such a conversion of unstructured text to the structured records will make it easy to retrieve, integrate with the Electronic-Health-Record (EHR) systems, and even with the advanced analytics to support prescription trends, dosage adherence, and drug interactions.

F. Preprocessing and Encoding

In case of additional computation operations, e.g. predictive modelling or classification, the structured data is first processed and coded into a numerical matrix format. This includes numerical normalization, one-hot representation of categorical features and feature scaling. Such measures help in making sure the information can be compatible with the machine learning models and minimize bias that comes about by unequal data distributions.

G. CNN-Based Feature Extraction

In the proposed system, Convolutional Neural Networks (CNNs) are utilized for image-based feature extraction during the region-of-interest (ROI) detection and text recognition stages. CNNs are well-suited for capturing spatial patterns and visual features from handwritten prescription images, enabling accurate localization and recognition of medicine-related regions. The current implementation focuses exclusively on CNN-based architectures integrated with YOLO for object detection and OCR for text extraction. Sequence-based models such as Long Short-Term Memory (LSTM) networks are not implemented in this work and are considered for future extensions to model sequential prescription patterns.

Convolutional Neural Networks (CNNs) are used in the suggested system solely to extract prescription image features and detect the region-of-interest inside the image. CNNs can be used to extract spatial and visual features of handwritten and printed medical text. Long Short-Term Memory (LSTM) models are sequences based and are not applied in the present work, as the task is to locate and recognize texts in the space, and not model sequences in time. Adopting LSTM-based architecture to estimate longitudinal prescription patterns is regarded as a possible future research addition.

V. RESULTS AND DISCUSSION

The current implementation focuses on prescription digitization and structured data generation. While the framework supports future predictive analytics, explicit disease prediction or clinical outcome forecasting models are not implemented in this study. Instead, the generated structured EHR data serves as a reliable foundation for subsequent predictive and risk assessment models. The experiment evaluation was done on a set of images of handwritten prescription that was obtained through various clinical sources. It contains more than 5000 prescription samples, different handwriting styles, prescription forms, and different image quality. The dataset was checked by hand so that the accuracy of annotation to detect ROI and OCR was verified.Fig. 4 presents an example of the system output showing successfully extracted and structured prescription information generated from a handwritten input image.In the current study, it is only the validation of prescription structure, the evaluation of the accuracy, and the benchmarking of the performance of OCR and detection modules, which are within the analytical capability of RxVision. Predictive analytics like disease progression or outcome prediction are not experimentally assessed but seen as future improvements made possible through the structure EHR representation generated by the system.The progressive performance improvement across each system stage demonstrates the soundness of the proposed validation methodology. The stepwise evaluation confirms the contribution of ROI detection, OCR enhancement, and spell correction to the final system accuracy. Fig.5 depicts the end-to-end pipeline from

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prescription image input through ROI detection, OCR, error correction, and structured EHR generation, enabling downstream analytical and risk assessment processes

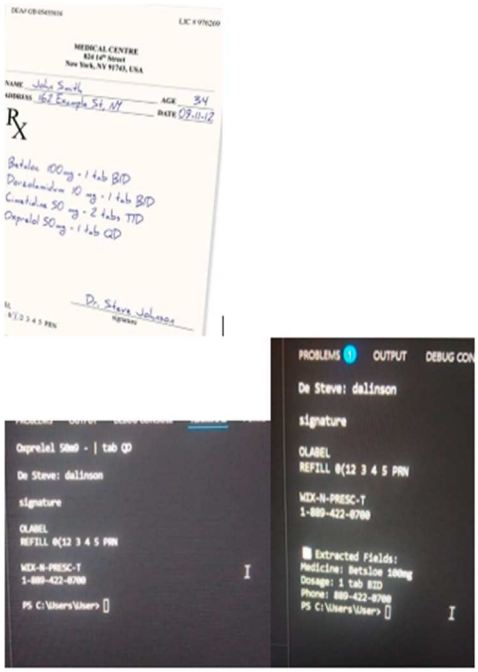
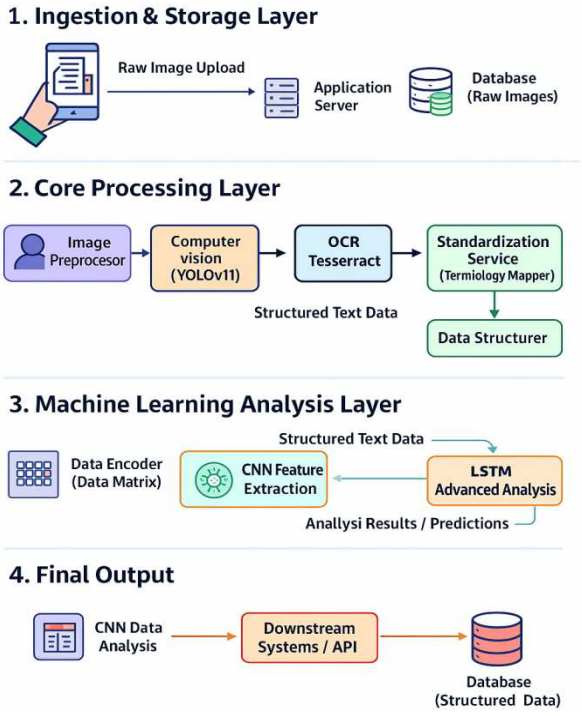


Fig.4.Sample output



. Fig.5. Workflow of Prescription to Structured EHR & Risk prediction

TABLE I. STAGE-WISE PERFORMANCE EVALUATION OF THE PROPOSED PRESCRIPTION ANALYSIS FRAMEWORK

Stage	*Description*	*Map@ 0.5*	*Precision (%)*	*Recall (%)*	F1- *Score*
Before enhancement	Baseline YOLO model	0.68	81.5	77.3	0.79
After YOLOv11	Added ROI detection for text area isolation	0.82	87.2	84.1	0.85
After OCR	Enhanced image contrast and noise removal before OCR	0.87	89.5	88.2	0.88
After Spell correction+ mapping	Applied TextBlob-based spell correction	0.90	92.4	89.6	0.91
Final integrated Model (YOLO +OCR+ Correction)	Final Integrated Model (YOLO+OCR+ Correction)	0.93	94.1	91.3	0.92

VI. CONCLUSION

RxVision presents an intelligent Electronic Health Record (EHR) framework designed to automate the analysis and digital transformation of handwritten prescriptions. The system employs YOLO-based Region of Interest (ROI) detection to accurately localize critical prescription sections, followed by OCR-based text extraction to convert handwritten or printed content into structured digital text. To ensure precision, CNN-driven feature encoding is integrated with an adaptive spell correction algorithm, minimizing transcription and recognition errors. This robust pipeline streamlines prescription digitization, improves data reliability, and make it seamlessly integrating it with the existing healthcare information systems. By incorporating both generic and brand medication mapping, RxVision enhances medication traceability and supports comprehensive clinical data management Overall performance was further enhanced by include spell correction in the finished model, which produced an F1-score of 0.92, a mAP@0.5 of 0.93, a precision of 94.1%, and a recall of 91.3%. This illustrates how the corrective layer successfully lowers errors brought on by OCR, leading to more precise and consistent data extraction.

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